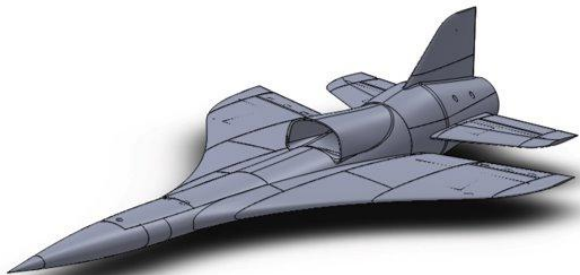


Demonstrator



Demonstrator Features

- Lightweight PLA Printed Airframe
- 1080mm Length / 600mm Wingspan
- 4 Servos (2 Flaperons & 2 Elevators)
- EDF with 20 N Thrust
- Flight Controller with Auto Stabilization
- **Target Speed: >150 mph**



Engine Overview



- **Basic Design Parameters**

- 450N Targeted Thrust (100lbf)
- ~150mm Diameter, ~400mm Length, ~4kg (5.9", 15.75", 8.82lb)
- Operating RPM: 100,000
- Air Mass Flow: .8kg/s (1.764lb/s)
- Expected Total Runtime: ~1 hour
- Expected Single Runtime: <7 minutes

- **Compressor**

- 1 Axial Stage, 1 Centrifugal Stage (6.44 OPR)

- **Combustor**

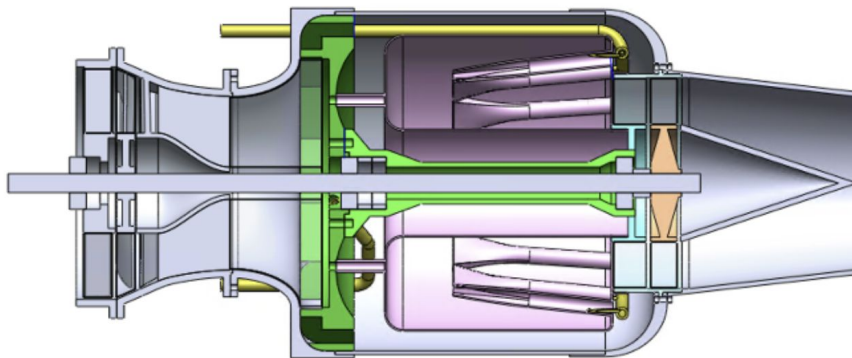
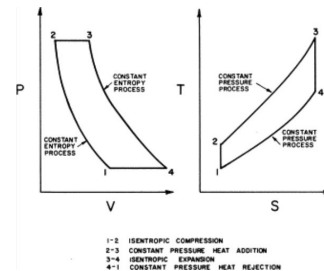
- Reverse Flow Annular Combustor
- Jet-A & Turbine Oil Mixture

- **Turbine**

- Single Stage
- Inconel 718

- **Nozzle**

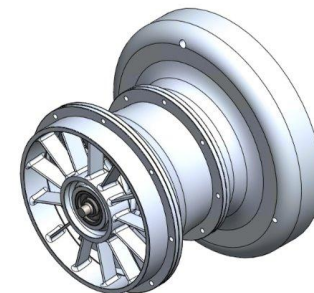
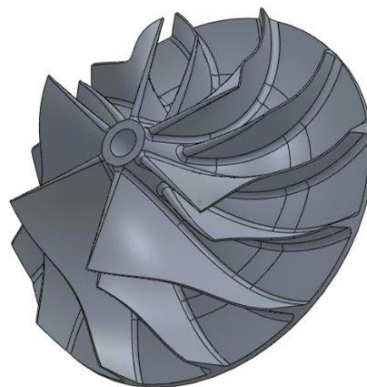
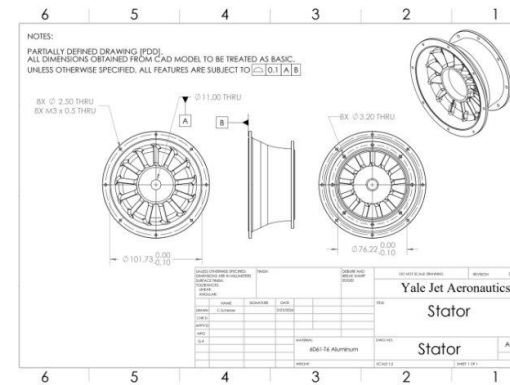
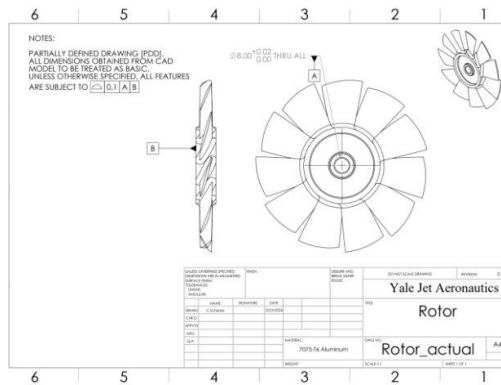
- Converging (Slightly Underexpanded)



Compressor



Design Parameters	
Quantity	Value
Axial Pressure Ratio	1.6
Centrifugal Pressure Ratio	4
Overall Pressure Ratio	6.44
Operating RPM	100K
Redline RPM	105K (0 FOS)



Compressor



- **Bearing Stack**

- **Compressor Testing Setup**

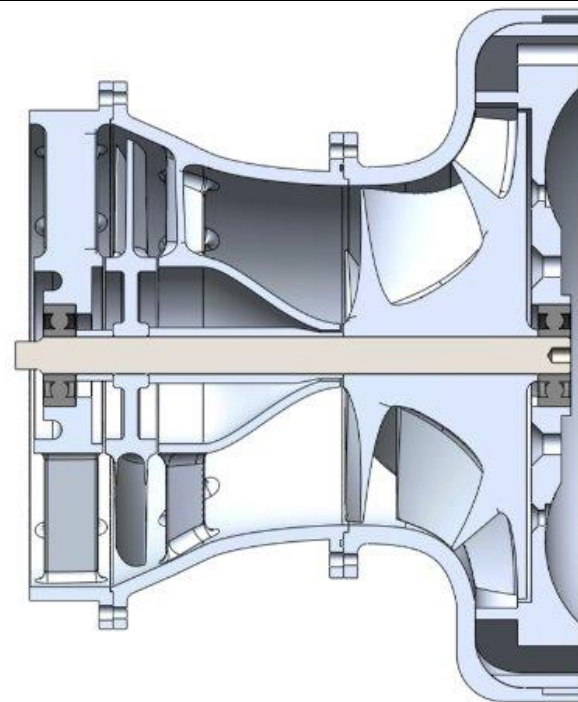
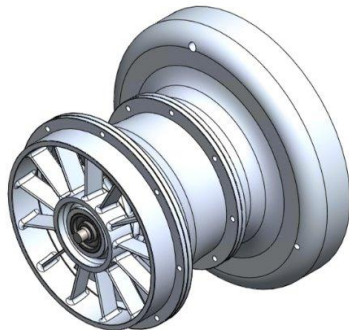
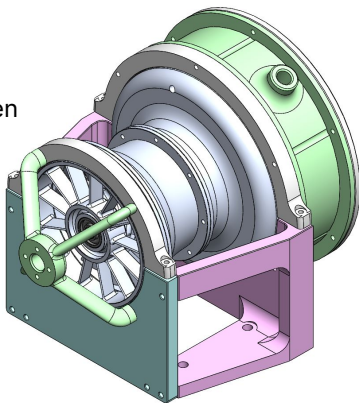
- Standard 608-2RS bearing

- **Engine Testing Setup**

- 608 silicon nitride mixed contact bearings (turbine and diffuser)
 - One ball bearing (axial compressor)

- H7/n6 transition fit for testing, interference for engine

Test enclosure:
PT access
Starter motor driven

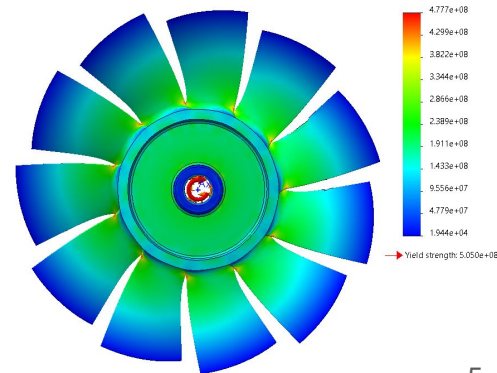
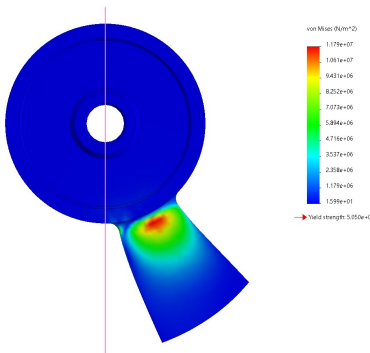
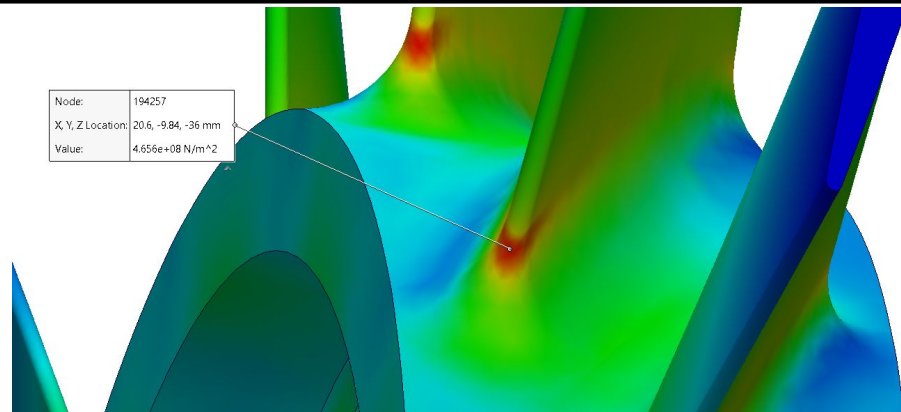


Compressor



- **Potential issue: Axial compressor centrifugal stress simulation**

- ~5% margin of safety, due to stress concentration at leading edge blade root.
- Production blades will have thicker roots, more tapered profile → Potential for efficiency drop
 - Consider transitioning from DCA to NACA 65 in order to increase mean thickness at root
- Passes 1.00kg/s mass flow tangential load with >30x FOS



Turbine



- **Geometric Considerations**

- Widen exit annulus to accommodate drop in density
- Blade aspect ratio of stator constrained by absolute Mach number out of the stator
- Designed to accelerate the gas axially
- Slight exit co-swirl
- Low degree of reaction: 0.104 → 0.461 (Hub → Tip)
potential flow separation concerns

===== SINGLE STAGE =====

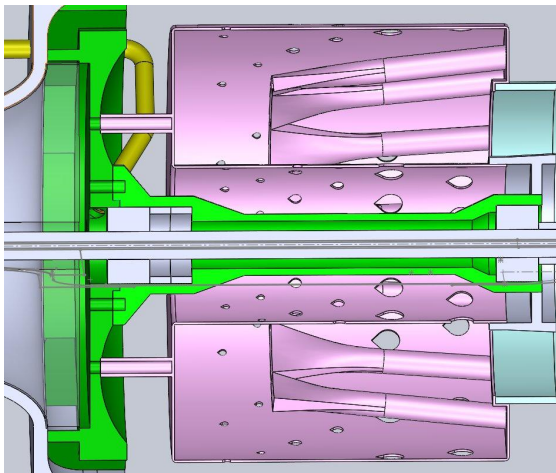
	Hub	Mean	Tip
r_in (mm)	38.0	44.0	50.0
r_exit (mm)	32.5	43.1	53.7
U_in (m/s)	397.9	460.8	523.6
U_exit (m/s)	340.1	451.2	562.4
c_2 (m/s)	734.6	639.7	565.5
c_3 (m/s)	246.3	213.4	161.8
c_theta_3 (m/s)	0.000	3.724	5.647
w_2 (m/s)	363.3	215.1	90.0
w_3 (m/s)	419.9	495.8	579.8
alpha_2 (deg)	75.5	77.4	81.6
alpha_3 (deg)	0.000	1.000	2.000
beta_2 (deg)	59.5	49.4	23.5
beta_3 (deg)	-54.1	-64.5	-73.8
Turning (deg)	113.6	113.9	97.3
Reaction	0.104	0.318	0.461
Delta_h (J/kg)	282144	285784	290176
M_c2	1.224	1.037	0.901
M_w2	0.606	0.349	0.143
M_w3	0.713	0.840	0.981

Combustor

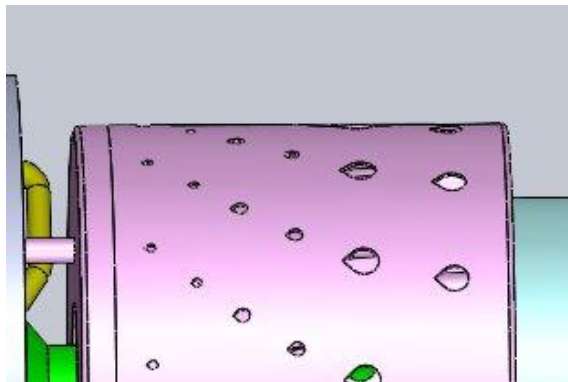


- Overall Design

- Material: Inconel 718
- 2 printed components fixed together by weld points
- Flame tubes mounted to the side of the chamber to allow high turbine radius
- 6% pressure drop across combustor liner
- AFR derived from T_4 value in similar microjet engines



OLD DESIGN - NOT REPRESENTATIVE



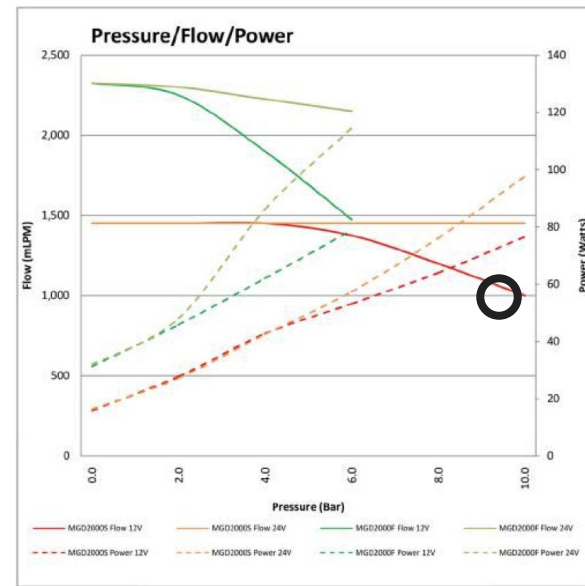
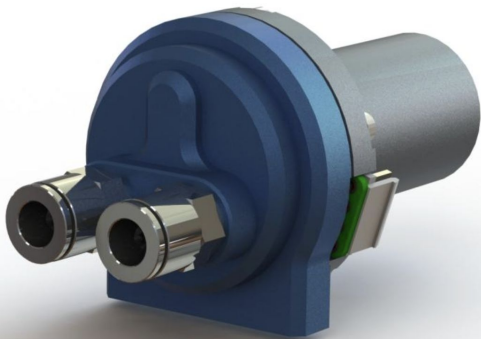
OLD DESIGN - NOT REPRESENTATIVE

Quantity	Value
T_{t3}	530 K (494 °F)
P_{t3}	6.84 atm (693063 Pa)
AFR	58.5
T_4	1180 K (1664 °F)
T_{t4}	1182 K (1668 °F)
P_{t4}	6.43 atm (651329 Pa)

Combustor



- **Fuel Pump Requirements**
 - Max Pressure: 9.5 bar derived from combustor pressure of 6.51 bar
 - Max Flow: 1000 ml/min
- **MGD2000S Fuel Pump**
 - Max Pressure: 10 bar
 - Max Free Flow: 1400 ml/min
 - Max Flow at Operating Pressure: 1100 ml/min



Nozzle



- **Design Methodology**
 - Slightly underexpanded to reduce complexity
 - Held dA/dx constant through exhaust cone region
 - Kept dA/dx similar through overhang region
- **Manufacturing**
 - Material: Stainless Steel 316
 - 2 printed components, mated by 3 M4 bolts
- **Seal**
 - Silver-Plated Inconel E-ring

